



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Thermal dissociation of a diatomic
 - To save some work, note that q_{trans} goes as $m^{3/2}$.
 - Take care with signs when evaluating $\Delta\epsilon_0$ in $\exp(-\Delta\epsilon_0 / kT)$
- Thermal ionization of atoms: the Saha equation
 - Ionization energy competes against entropy of the free electron.
- Adsorption of a gas on a surface: the Langmuir isotherm
 - Equilibrium condition: $\mu_{\text{gas}} = \mu_{\text{adsorbed}}$
 - States available to an adsorbed atom depend on coverage of the surface.
 - Must go back to the system partition function Q_N .
 - Surface sites are distinguishable, but molecules are not.
 - Thermodynamic expression for the coverage θ gives physical insight.

Summary of last lecture

- Potential energy surfaces
 - The PES is the electronic energy as a function of the nuclear positions.
 - Stable molecules are minima on the PES.
 - Transition state:
 - The highest point on the lowest-energy path
 - Maximum along the reaction coordinate, minimum in all other directions

Transition state theory

- The fraction of molecules coming together with sufficient energy to reach the transition state is given by **$\exp(-E_a/RT)$** .
- For a typical value of the activation energy of 50 kJ mol⁻¹, this means that *only about 1 in 10⁹ of the molecules actually have enough energy to make it.*
- We can imagine the molecules all queuing up like eager climbers at Everest base camp (the reactants); many set off towards the summit (the transition state), but few have the energy to make it, and so slide back down the slope.

Transition state theory

- *This rarity of reaching the transition state leads to the idea that it is possible to assume that the transition states are in equilibrium with the reactants.* The argument is that there is much toing and froing up the slope that a proper equilibrium is established. In addition, once formed the transition state does not immediately go to products – rather the process has a finite rate, meaning that *the transition state has a finite lifetime.*
- There is therefore the possibility that *the transition state will return to reactants before going on to products, thereby establishing the equilibrium with the reactants.* The kinetic scheme is:



Transition state theory



- Assuming equilibrium between A, BC and TS, we may write

$$K^* = \frac{[TS] c^\circ}{[A] [BC]}$$

where K^ is the equilibrium constant for the formation of the transition state, $[TS]$ is the concentration of the transition states and so on. Recall from the discussion in last section that *the factor of c° , the standard concentration, is needed in order to make the equilibrium constant dimensionless.**

Transition state theory

- The rate, r , at which the transition state breaks down to form products is assumed to be a simple first order process

$$r = k_{1\text{st}} [\text{TS}]$$

so

$$r = k_{1\text{st}} K^* \frac{[\text{A}][\text{BC}]}{c^\circ}$$

The reaction is bimolecular and so its rate law will be

$$r = k_{2\text{nd}} [\text{A}][\text{BC}].$$

Comparing the two expressions of the rate, r , we can find the following expression for the second order rate constant, $k_{2\text{nd}}$:

$$k_{2\text{nd}} = \frac{1}{c^\circ} k_{1\text{st}} K^*.$$

The problem reduces to the calculation of $k_{1\text{st}}$ and K^ , and we shall see that this can be done using statistical thermodynamics.*

Calculation of $K^\ddagger$ and k_{1st}

- To compute the equilibrium constant for the formation of the transition states we can use the equation:

$$K_c = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \left(\frac{1}{c^\circ} \right)^{\Delta \nu} \exp(-\Delta \epsilon_0 / kT)$$

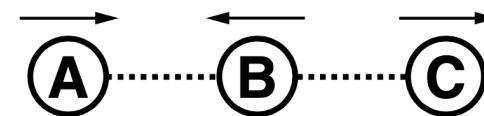
- In the present case, this becomes

$$K^* = \frac{f_{TS}}{f_A f_{BC}} \left(\frac{1}{c^\circ} \right)^{-1} \exp(-\Delta \epsilon_0^\ddagger / kT)$$

where $\Delta \epsilon_0^\ddagger$ is now the difference in energy between the electronic ground states of the reactants, A and BC, and the transition state.

Calculation of $K^\ddagger$ and k_{1st}

- *When computing the vibrational contribution to f_{TS} we realize that one of the vibrational normal modes of the TS corresponds to motion along the reaction co-ordinate.* In the case of the linear A + BC encounter, this normal mode is a kind of asymmetric stretch in which the A–B distance decreases and the B–C distance increases. If pushed far enough this normal mode will lead to the transition state breaking down to give products. The other crucial thing about this normal mode is that as it takes place at a potential energy maximum; there is no restoring force. So, this vibration is very unusual.



The asymmetric vibration of the transition state, in which the AB distance decreases and the BC distance increases, is different from the other normal modes in that it is the motion which leads to products, AB + C.

Calculation of $K^\ddagger$ and k_{1st}

- We will therefore factor the partition function for vibration along the reaction co-ordinate out of f_{TS} :

$$K^* = \frac{q^\ddagger f'_{TS}}{f_A f_{BC}} \left(\frac{1}{c^\circ} \right)^{-1} \exp(-\Delta\varepsilon_0/kT)$$

where $q^\ddagger$ is the partition function for vibration along the reaction co-ordinate and f'_{TS} is the partition function for the transition state omitting the contribution from the vibration along the reaction co-ordinate.

Calculation of $K^\ddagger$ and k_{1st}

- The partition function $q^\ddagger$ can be written:

$$q = \frac{\exp(-h\nu^\ddagger/2kT)}{1 - \exp(-h\nu^\ddagger/kT)}$$

where $\nu^\ddagger$ is the vibrational frequency of this special normal mode.

- We then recognize that *as this vibration actually leads the transition state to dissociate the frequency $\nu^\ddagger$ is likely to be low.* Assuming that it is so low that $h\nu^\ddagger/kT \ll 1$ we can approximate the exponentials and hence simplify the expression for the partition function:

$$\frac{\exp(-h\nu^\ddagger/2kT)}{1 - \exp(-h\nu^\ddagger/kT)} \approx \frac{1 - h\nu^\ddagger/2kT}{1 - 1 + h\nu^\ddagger/kT} = \frac{kT}{h\nu^\ddagger}.$$

- So our final expression for K^* is

$$K^* = \underbrace{\frac{kT}{h\nu^\ddagger}}_{q^\ddagger} \underbrace{\frac{f'_{TS}}{f_A f_{BC}} \left(\frac{1}{c^\circ}\right)^{-1} \exp(-\Delta\varepsilon_0^\ddagger/kT)}_{K^\ddagger}$$

Calculation of $K^\ddagger$ and k_{1st}

- The first order rate constant, k_{1st} , for the decay of the transition state to product can now be identified with the frequency of the vibration along the reaction co-ordinate, $\nu^\ddagger$. The argument is that *each oscillation along this co-ordinate leads to the transition state breaking apart, so the rate constant for this first order process is the frequency of oscillation:*

$$k_{1st} = \nu^\ddagger$$

Calculation of $K^\ddagger$ and $k_{1\text{st}}$

- So

$$\begin{aligned}k_{2\text{nd}} &= \frac{1}{c^\circ} k_{1\text{st}} K^\ddagger \\&= \frac{1}{c^\circ} \nu^\ddagger \frac{kT}{h\nu^\ddagger} \frac{f'_{\text{TS}}}{f_{\text{A}}f_{\text{BC}}} \left(\frac{1}{c^\circ}\right)^{-1} \exp(-\Delta\varepsilon_0^\ddagger/kT) \\&= \frac{kT}{h} \frac{f'_{\text{TS}}}{f_{\text{A}}f_{\text{BC}}} \exp(-\Delta\varepsilon_0^\ddagger/kT)\end{aligned}$$

where, conveniently, the frequency $\nu^\ddagger$ has cancelled. Recall that the units of f are (in SI) m^{-3} and that (kT/h) will have units s^{-1} , so the RHS (the right-hand-side of equation) has units of $\text{m}^3 \text{s}^{-1}$ which is correct for a second order rate constant in molecular units. If we wanted the rate constant in $\text{mole}^{-1} \text{m}^3 \text{s}^{-1}$ we would have to multiply by Avogadro's number, N_{A} .

- *It is easy to get confused in this section between Boltzmann's constant, k , and various rate constants. The symbol k is always Boltzmann's constant; rate constants are distinguished from it by subscripts e.g. $k_{2\text{nd}}$.*

Applications of transition state theory

$$k_{2\text{nd}} = \frac{kT}{h} \frac{f'_{\text{TS}}}{f_{\text{A}}f_{\text{BC}}} \exp\left(-\Delta\epsilon_0^\ddagger/kT\right)$$

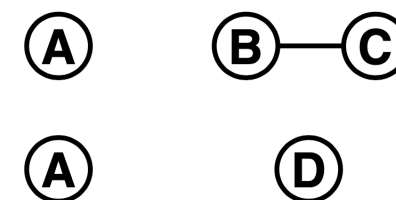
- This equation looks very appealing as it appears to give us a way of computing the rate constant from a knowledge of molecular energy levels, just as we have done for thermodynamic parameters and equilibrium constants.
- However, to use it we need to know about the energy levels of the transition state, as these are needed to find f'_{TS} . *This is a great problem as the transition state only has a fleeting existence (typically 10^{-13}s) and so is not amenable to study.* We also need to know $\Delta\epsilon_0^\ddagger$, the difference in energy between the electronic ground states of the reactants and the transition states; again, *this is not experimentally accessible.*

Applications of transition state theory

- All is not quite as bad as it seems, though. *The first possibility is to use the PES to find out about the transition state.* The reaction co-ordinate and the position of the transition state can be identified from this surface, and this would allow us to determine both the geometry and energy of the transition state. *However, computing such a detailed and accurate PES for even a simple reaction remains a formidable task with present computing resources.* Also, if such a detailed PES is available, more complex theories of reaction rates (such as *molecular reaction dynamics*) would be used in preference to transition state theory.
- Faced with these problems it is clear that we will not be able to calculate absolute values of rate constants 'on the back of an envelope'. What we can do, however, is to use the theory to make some qualitative predictions and rationalisations.

Steric factors

- Consider once more the $A + BC$ reaction with a linear geometry. The transition state looks rather like a linear tri-atomic $A-B-C$, although we do not know what the relative bond lengths are. *By assuming that the attack of A has to be along the line $B-C$ we have introduced a steric requirement into this reaction. If, on the other hand, we imagined that BC was structureless (say an atom D), then there would be no steric requirement.* What we are going to do is to estimate the rate constants in these two cases and then compare them to find a value for the steric factor.



The reaction between A and BC has a steric requirement as an encounter is more likely to be successful if A approaches B along the line BC. If we suppose that BC has no structure, modelling it as an atom D, the steric requirement is removed.

Steric factors

- Firstly we note that partition functions from translation, rotation and vibration are very different in size, with $q_{\text{trans}} \gg q_{\text{rot}} \gg q_{\text{vib}}$. Given that this is so, we can say that *to some approximation the thing that is important is not the actual values of each contribution but its type i.e. translation, rotation or vibration.*
- We will write the contribution to the partition function from a translational degree of freedom as **{trans}**, from a rotational degree of freedom as **{rot}** and from a vibrational degree of freedom as **{vib}**. These individual contributions are assumed to be the same for all molecules.
- Using this approach the partition function for the diatomic BC is estimated as

$$f_{\text{BC}} = \{\text{trans}\}^3 \{\text{rot}\}^2 \{\text{vib}\}$$

- *Note that there are three translational degrees of freedom (x, y and z) and two rotational degrees of rotational freedom for a diatomic.* For A, an atom, there are simply three translational degrees of freedom:

$$f_{\text{A}} = \{\text{trans}\}^3$$

Steric factors

- For the linear transition state there are two rotational contributions (as the diatomic). A linear tri-atomic has **(3N – 5)** vibrational modes i.e. 4, but remember that *one of these has been factored out as it corresponds to vibration along the reaction co-ordinate*. So

$$f'_{TS} = \{\text{trans}\}^3 \{\text{rot}\}^2 \{\text{vib}\}^3$$

- Putting this all together in the equation

$$k_{2\text{nd}} = \frac{kT}{h} \frac{f'_{TS}}{f_A f_{BC}} \exp(-\Delta\epsilon_0^\ddagger/kT)$$

- gives

$$\begin{aligned} k_{2\text{nd}} (A + BC) &= \frac{kT}{h} \frac{\{\text{trans}\}^3 \{\text{rot}\}^2 \{\text{vib}\}^3}{\{\text{trans}\}^3 \{\text{trans}\}^3 \{\text{rot}\}^2 \{\text{vib}\}} \exp(-\Delta\epsilon_0^\ddagger/kT) \\ &= \frac{kT}{h} \frac{\{\text{vib}\}^2}{\{\text{trans}\}^3} \exp(-\Delta\epsilon_0^\ddagger/kT) \end{aligned}$$

Steric factors

- The table below shows that the number and different types of degrees of freedom available to the reactants and the transition state:

type	A	BC	(A + BC)	TS
trans.	3	3	6	3
rot.	0	2	2	2
vib.	0	1	1	3

- We see that in forming the TS there is a loss of three translational degrees of freedom; this has a very significant effect of the value of k_{2nd} as the corresponding partition functions are large. There is a gain of two vibrational degrees of freedom, but the corresponding partition functions are small so this does not make such large change to k_{2nd} .

Steric factors

- The calculation is now repeated assuming that BC is structureless i.e. an atom D, so that

$$f_D = \{\text{trans}\}^3$$

- The transition state is now effectively a diatomic and so the one vibrational degree of freedom corresponds to motion along the reaction co-ordinate. Therefore

$$f'_{\text{TS}} = \{\text{trans}\}^3 \{\text{rot}\}^2$$

- So, the rate constant is now

$$\begin{aligned} k_{2\text{nd}}(\text{A} + \text{D}) &= \frac{kT}{h} \frac{\{\text{trans}\}^3 \{\text{rot}\}^2}{\{\text{trans}\}^3 \{\text{trans}\}^3} \exp(-\Delta\epsilon_0^\ddagger/kT) \\ &= \frac{kT}{h} \frac{\{\text{rot}\}^2}{\{\text{trans}\}^3} \exp(-\Delta\epsilon_0^\ddagger/kT). \end{aligned}$$

Steric factors

- The table looks somewhat different:

type	A	D	(A + D)	TS
trans.	3	3	6	3
rot.	0	0	0	2
vib.	0	0	0	0

- Again there is a loss of three translational degrees of freedom. However, two rotational degrees of freedom are gained, and this is more significant than a gain in vibrational degrees of freedom as the rotational partition functions are larger.

Steric factors

$$k_{2\text{nd}}(\text{A} + \text{BC}) = \frac{kT}{h} \frac{\{\text{vib}\}^2}{\{\text{trans}\}^3} \exp(-\Delta\varepsilon_0^\ddagger/kT)$$

$$k_{2\text{nd}}(\text{A} + \text{D}) = \frac{kT}{h} \frac{\{\text{rot}\}^2}{\{\text{trans}\}^3} \exp(-\Delta\varepsilon_0^\ddagger/kT)$$

- The ratio of the two rate constants gives the steric factor

$$p = \frac{k_{2\text{nd}}(\text{A} + \text{BC})}{k_{2\text{nd}}(\text{A} + \text{D})} = \frac{\{\text{vib}\}^2}{\{\text{trans}\}^3} \frac{\{\text{trans}\}^3}{\{\text{rot}\}^2} = \frac{\{\text{vib}\}^2}{\{\text{rot}\}^2}$$

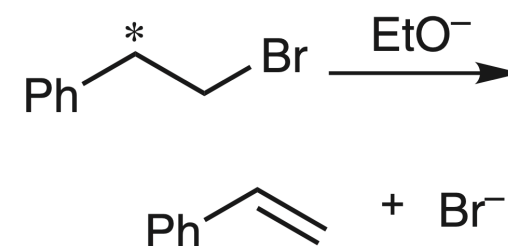
- In next example we will see that using this expression ***p*** is predicted to be very much less than 1. Our interpretation of this is that when the reactant BC has structure the value of the equilibrium constant ***K*[‡]** is reduced when compared to the case where BC is considered to be structureless.

Example

- Estimate the steric factor at 298 K for the $A + BC$ reaction proceeding via a linear transition state.

Kinetic isotope effects

- For some reactions it is observed that the substitution of one isotope for another causes a significant change in the rate constant; this is called the *kinetic isotope effect, KIE*.
- The most dramatic changes are seen when deuterium is substituted for hydrogen; in such cases a change by a factor of between 3 and 7 is not unusual. The implication of an observation of a large isotope effect is that the substituted atom must be intimately involved in the reaction.
- *KIEs are used to provide information about the mechanisms of organic reactions.* All we are going to do here is to use transition state theory to provide an estimate of the KIE in a simple case.



Substitution of H by D at position * in this elimination reaction causes a reduction in the rate constant by a factor of approx 7.

Kinetic isotope effects

- *We know that simply substituting one isotope for another cannot alter the electronic structure of the molecule as the nuclear charges and the number of electrons remain the same.*
- For large molecules, substituting D for H is unlikely to alter the moment of inertia to a significant extent. However, what will alter are the vibrational frequencies of normal modes which involve the substituted atom. Recall from previous section that, using the simple harmonic oscillator model, we predict that the ratio of the frequencies depends on the reduced masses according to

$$\frac{\nu_A}{\nu_B} = \sqrt{\frac{\mu_B}{\mu_A}}$$

where A and B denote isotopic species. *It is in this effect that the KIE originates.*

Kinetic isotope effects

- Suppose that we have a reaction in which a bond to an H atom is broken. As discussed before the only contributions to the partition function of the reactants which will be different are those from the vibrational modes involving the H (or D). Let us assume that there is one such mode, whose partition function can be written

$$q_{\text{vib,H}} = \frac{\exp(-h\nu_{\text{H}}/2kT)}{1 - \exp(-h\nu_{\text{H}}/kT)}$$

- A similar expression can be written for the case where the proton has been replaced by deuterium:

$$q_{\text{vib,D}} = \frac{\exp(-h\nu_{\text{D}}/2kT)}{1 - \exp(-h\nu_{\text{D}}/kT)}$$

Kinetic isotope effects

$$q_{\text{vib,H}} = \frac{\exp(-h\nu_{\text{H}}/2kT)}{1 - \exp(-h\nu_{\text{H}}/kT)} \quad q_{\text{vib,D}} = \frac{\exp(-h\nu_{\text{D}}/2kT)}{1 - \exp(-h\nu_{\text{D}}/kT)}$$

- The frequencies of these vibrations are likely to be high as a light atom is involved, in which case **$h\nu \gg kT$** . We can therefore approximate **$1 - \exp(-h\nu_{\text{H}}/kT) \approx 1$** to give:

$$q_{\text{vib,H}} \approx \exp(-h\nu_{\text{H}}/2kT) \quad q_{\text{vib,D}} \approx \exp(-h\nu_{\text{D}}/2kT)$$

- Recall that these terms **$\exp(-h\nu/2kT)$** arise from the zero point energy (ZPE) of the oscillator. So what we are saying by making these approximations is that *the only the $v = 0$ level contributes to the partition function.*

Kinetic isotope effects

- In the transition state the vibration involving the H (or D) is the one which corresponds to the reaction co-ordinate and so is factored out of the problem. Also, $\Delta\epsilon_0^\ddagger$ is the same for isotopic species as such substitutions do not affect the electronic energies.
- In computing the ratio of the rate constants for the substituted molecules, all of the factors in the equation of $k_{2\text{nd}}$ cancel except for those due to the vibration involving H (or D):

$$k_{2\text{nd}} = \frac{kT}{h} \frac{f'_{\text{TS}}}{f_{\text{A}}f_{\text{BC}}} \exp(-\Delta\epsilon_0^\ddagger/kT)$$

$$\frac{k_{2\text{nd}}(\text{H})}{k_{2\text{nd}}(\text{D})} = \frac{q_{\text{vib,D}}}{q_{\text{vib,H}}} = \frac{\exp(-h\nu_{\text{D}}/2kT)}{\exp(-h\nu_{\text{H}}/2kT)} = \exp\left(\frac{-h(\nu_{\text{D}} - \nu_{\text{H}})}{2kT}\right)$$

- Recall that the ZPE is $1/2h\nu$, so this expression can be written as

$$\frac{k_{2\text{nd}}(\text{H})}{k_{2\text{nd}}(\text{D})} = \exp\left(\frac{-(\text{ZPE}_{\text{D}} - \text{ZPE}_{\text{H}})}{kT}\right)$$

- *The rate constants differ by a factor which depends on the difference of the vibrational zero point energies of the two species.*

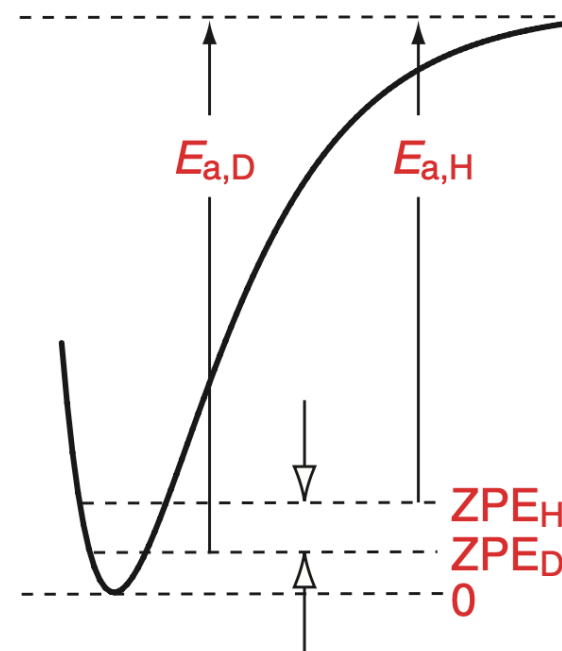
Kinetic isotope effects

- A simple interpretation of the KIE is that the activation energy for the deuterated species is higher than for the protonated species on account of the lower ZPE of the deuterated species e.g. we imagine the reaction involving reactants in the lowest vibrational level. The difference in the activation energy is therefore the difference in the ZPEs between the two species, which leads to the expression given before for the ratio of the rate constants.

$$k_{2\text{nd}}(\text{H}) = A \exp(-E_{\text{a,H}}/RT) \quad k_{2\text{nd}}(\text{D}) = A \exp(-E_{\text{a,D}}/RT)$$

hence

$$\begin{aligned} \frac{k_{2\text{nd}}(\text{H})}{k_{2\text{nd}}(\text{D})} &= \exp\left(\frac{-E_{\text{a,H}} + E_{\text{a,D}}}{kT}\right) \\ &= \exp\left(\frac{\text{ZPE}_{\text{H}} - \text{ZPE}_{\text{D}}}{kT}\right) \end{aligned} \quad \text{by looking at the figure}$$



Example

- Estimate the size of the kinetic isotope effect expected at 298 K when D is substituted for H.

Thermodynamic formulation of transition state theory

- As we have seen, *the problem with transition state theory is that although it offers the possibility of calculating rate constants ab initio, in practice such calculations are rather difficult.*
- An alternative approach is to utilise the basic ideas of the theory but express them in the language of classical thermodynamics. This leads to a formulation of the problem which is not predictive but is very useful for thinking about the factors which influence rate constants, especially for reactions in solution.

Thermodynamic formulation of transition state theory

- We start from the equation:

$$k_{2\text{nd}} = \frac{1}{c^\circ} k_{1\text{st}} K^*$$

- And, K^* can be rewritten

$$K^* = q^\ddagger K^\ddagger$$

- To a good approximation, the partition function for vibration along the reaction coordinate, $q^\ddagger$, is given by

$$q^\ddagger = \frac{kT}{h\nu^\ddagger}$$

- Putting these all together, the expression for $k_{2\text{nd}}$ becomes

$$k_{2\text{nd}} = \frac{1}{c^\circ} k_{1\text{st}} \frac{kT}{h\nu^\ddagger} K^\ddagger$$

- As before, we conveniently identify $k_{1\text{st}}$ with $\nu^\ddagger$ to give what is known as the Eyring equation

$$k_{2\text{nd}} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger.$$

Thermodynamic formulation of transition state theory

- The next step is to use classical thermodynamics to write the equilibrium constant in terms of $\Delta_r G^\circ$:

$$\Delta_r G^{\circ,\ddagger} = -RT \ln K^\ddagger$$

where $\Delta_r G^{\circ,\ddagger}$ is the change in Gibbs energy, under standard conditions, when the transition state is formed from the reactants.

- *We have to be very careful in using this expression as it assumes that the same standard state is used in both the definition of $K^\ddagger$ and of $\Delta_r G^{\circ,\ddagger}$.* Normally, the standard state chosen for $\Delta_r G^\circ$ is, for gases, a pressure of 1 bar and, for solutions, a concentration of 1 mol dm⁻³; however, we have so far defined $K^\ddagger$ using a standard state of 1 molecule m⁻³, which is not the usual standard state for $\Delta_r G^\circ$.
- Sorting out this difficulty is just a question of a change of standard states.

Thermodynamic formulation of transition state theory

- So for a gas phase reaction, is

$$\text{gas-phase} \quad k_{2\text{nd}} / (\text{N m}^{-2})^{-1} \text{ s}^{-1} = \left(\frac{1}{p^\circ} \right) \frac{kT}{h} \exp \left(\frac{-\Delta_r G^{\circ,\ddagger}}{RT} \right)$$

where p° is the standard pressure of 1 bar or 10^5 N m^{-2} , and $\Delta_r G^{\circ,\ddagger}$ is referred to this standard state. *Note the units of $k_{2\text{nd}}$.*

- For a reaction in solution the result has a similar form

$$\text{solution} \quad k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \left(\frac{1}{[\]^\circ} \right) \frac{kT}{h} \exp \left(\frac{-\Delta_r G^{\circ,\ddagger}}{RT} \right)$$

where $[\]^\circ$ is the standard concentration of 1 mol dm^{-3} , and $\Delta_r G^{\circ,\ddagger}$ is referred to this standard state. Once again, note the units of $k_{2\text{nd}}$.

Thermodynamic formulation of transition state theory

- The standard Gibbs energy change can be written in terms of a standard enthalpy change and a standard entropy change:

$$\Delta_r G^{\circ, \ddagger} = \Delta_r H^{\circ, \ddagger} - T \Delta_r S^{\circ, \ddagger}$$

- Using this, the expression for the rate constant in solution becomes

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^{\circ}} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ, \ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ, \ddagger}}{RT}\right)$$

- *This is the equation which we use to interpret rate constants in this thermodynamic formulation.*

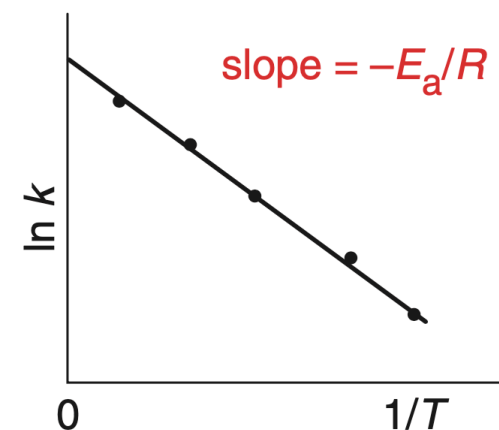
Relationship to the Arrhenius equation

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right)$$

- This equation is very reminiscent of the Arrhenius equation:

$$k_{2\text{nd}} = A \exp\left(\frac{-E_a}{RT}\right)$$

- Tempting though it is to identify E_a as $\Delta_r H^{\circ,\ddagger}$, this would be wrong because it does not take account of the formal definition of the activation energy, which is as follows. *The slope (at a particular temperature) of a plot of $\ln k_{2\text{nd}}$ against $1/T$ has the value $-E_a/R$; this defines the activation energy.* In addition, the rate constants must be recorded at constant volume.



The activation energy is defined from the slope of a plot of $\ln k$ against $1/T$; the slope may vary as a function of temperature.

Relationship to the Arrhenius equation

- In the language of calculus this definition becomes

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial (1/T)} \right)_V = -\frac{E_a}{R}$$

or, entirely equivalently,

$$\boxed{\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{E_a}{RT^2}.$$

- The simplest way to find E_a in terms of the other parameters used in transition state theory is to differentiate the equation of $k_{2\text{nd}}$

$$\boxed{k_{2\text{nd}} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger.$$

first having taken logarithms of both sides

$$k_{2\text{nd}} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger$$

$$\ln k_{2\text{nd}} = \ln \frac{k}{c^\circ h} + \ln T + \ln K^\ddagger$$

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{1}{T} + \left(\frac{\partial \ln K^\ddagger}{\partial T} \right)_V$$

Reactions in solution

- The variation of the equilibrium constant with temperature is given by the familiar van't Hoff isochore

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\circ}{RT^2}$$

- Reactions in solution are little affected by changes in pressure, so we can safely use this relationship to substitute for $(\partial \ln K^\ddagger / \partial T)_V$ to give

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{1}{T} + \left(\frac{\partial \ln K^\ddagger}{\partial T} \right)_V = \frac{1}{T} + \frac{\Delta_r H^{\circ, \ddagger}}{RT^2}$$

- Comparing this to the equation

$$\boxed{\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{E_a}{RT^2}}$$

we have

$$\frac{E_a}{RT^2} = \frac{1}{T} + \frac{\Delta_r H^{\circ, \ddagger}}{RT^2}$$

$$\text{hence } E_a = \Delta_r H^{\circ, \ddagger} + RT \quad (\text{in solution})$$

Gas phase reactions

- For gas phase reactions the situation is a little more complex as the van't Hoff isochore applies only to the equilibrium constant expressed in terms of partial pressures, K_p :

$$\frac{d \ln K_p}{dT} = \frac{\Delta_r H^\circ}{RT^2}$$

- Recall that $K^\ddagger$ is expressed in terms of concentrations, specifically in *molecules per unit volume*. The relationship between K_p and the equilibrium constant in terms of concentrations, K_c , is easily worked out for a particular reaction:



$$K_p = \frac{p_C p^\circ}{p_A p_B}$$

$$\text{but } p_i = \frac{n_i RT}{V} = \frac{N_i kT}{V} = c_i kT$$

$$\text{so } K_p = \left(\frac{c_C}{c_A c_B} \right) \frac{p^\circ}{c^\circ kT} = K_c \frac{p^\circ}{c^\circ kT}$$

Gas phase reactions

$$K_p = \left(\frac{c_C}{c_A c_B} \right) \frac{p^\circ}{c^\circ kT} = K_c \frac{p^\circ}{c^\circ kT}$$

- where for species i p_i is the partial pressure, n_i is the number of moles, N_i is the number of molecules and c_i is the concentration in molecules per unit volume; we have also used $R = N_A k$. So, substituting this expression for K_p into the van't Hoff isochore we have

$$\begin{aligned} \frac{d}{dT} \left(\ln K_c - \ln T + \ln \frac{p^\circ}{kc^\circ} \right) &= \frac{\Delta_r H^\circ}{RT^2} \\ \frac{d \ln K_c}{dT} - \frac{1}{T} &= \frac{\Delta_r H^\circ}{RT^2} \\ \frac{d \ln K_c}{dT} &= \frac{\Delta_r H^\circ + RT}{RT^2} \end{aligned}$$

- Then using this we can have

$$\begin{aligned} \left(\frac{\partial \ln k_{2nd}}{\partial T} \right)_V &= \frac{1}{T} + \left(\frac{\partial \ln K^\ddagger}{\partial T} \right)_V \\ &= \frac{1}{T} + \frac{\Delta_r H^{\circ, \ddagger} + RT}{RT^2} \\ &= \frac{\Delta_r H^{\circ, \ddagger} + 2RT}{RT^2}. \end{aligned}$$

Gas phase reactions

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{E_a}{RT^2}.$$

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{\Delta_r H^{\circ, \ddagger} + 2RT}{RT^2}.$$

- Comparing the two equations we see that for the bimolecular gas phase reaction

$$E_a = \Delta_r H^{\circ, \ddagger} + 2RT \quad (\text{bimolecular, gas phase}).$$

- Admittedly, this discussion is rather convoluted and leads to no startling conclusion. It is important, however, to realise that the enthalpy of activation is not the same as the activation energy.

The pre-exponential or A-factor

- Returning to the equation

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right)$$

and substituting in for $\Delta_r H^{\circ,\ddagger}$ we have, for a reaction in solution,

$$\begin{aligned} k_{2\text{nd}} &= \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-E_a + RT}{RT}\right) \\ &= \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) e^1 \exp\left(\frac{-E_a}{RT}\right) \\ &= \left[\frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R} + 1\right) \right] \exp\left(\frac{-E_a}{RT}\right) \end{aligned}$$

where, for convenience, we have put the factor of e into the first exponential. By comparison with the Arrhenius equation, the quantity in the square bracket can be equated with the pre-exponential or **A**-factor. *We see that the entropy of activation is the significant contributor to the A-factor.*